WADIH G. DAHROUGE

Robotics & Autonomous Systems Engineer | UAV Navigation, Perception & Multi-Sensor Fusion

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PROFILE & RESEARCH INTERESTS

Mechatronics engineer with a research focus on UAV autonomy, GNSS-degraded navigation, multi-sensor fusion, and deep-learning-based perception. Holds a 5-year Bachelor of Engineering (155 US credit hours / ~300 ECTS), structurally equivalent to a Bachelor+Master (M2), EQF Level 7, from La Sagesse University, Beirut. Led the technical design and build of a fully autonomous disaster-reconnaissance UAV as FYP Technical Lead — self-funded, awarded A+, and formally recognized by the Lebanese Civil Defence as operationally indispensable. Seeking a fully-funded PhD position to develop robust state-estimation and navigation methods for aerial robots operating in GNSS-denied or perceptually degraded environments. Industry-trained in PLC/VFD automation and photonics assembly at Multilane Inc. Trilingual (Arabic, English, French), CCNA certified.

EDUCATION

Bachelor of Engineering in Mechatronics (5-year, ~300 ECTS) *2019 – 2025*

La Sagesse University (ULS), Beirut, Lebanon

- Equivalent to M2 level under the Bologna Process (EQF Level 7).
- Final Year Project: A+ — Autonomous UAV for Disaster Reconnaissance (Technical Lead).

FLAGSHIP RESEARCH PROJECT — FINAL YEAR PROJECT (A+)

Autonomous UAV System for Rapid Disaster Reconnaissance

La Sagesse University | Grade: A+ | Technical Lead | Proposed for operational deployment — Lebanese Civil Defence

- As Technical Lead of a 3-person team, carried out the majority of the system's design, integration, and implementation across the full autonomy stack — mechanical frame (3D CAD), electronics stack, and software pipeline — and personally covered the majority of project costs; the entire system was self-funded with no institutional support.
- Implemented a ROS2-based autonomy architecture bridged to ArduPilot/Pixhawk via MAVROS, covering flight control, state management, and inter-module communication.
- Integrated a 360° LiDAR pipeline for real-time obstacle detection and avoidance, validated in RViz simulation and open-field flight tests to ±2 cm tolerance against ground-truth measurements.
- Developed a multi-sensor fusion pipeline combining camera, IMU, and range sensors for robust state estimation and situational awareness (GPS accuracy ±2–3 m in field tests) — the core estimation layer underpinning the navigation stack, and the direction this research interest builds on.
- Trained and deployed deep-learning computer vision models on a companion computer (Raspberry Pi) for real-time detection: 92% human detection accuracy (YOLOv8) and 80% fire detection accuracy.
- Validated the system across bench-level checks, QGroundControl/Mission Planner mission simulation, and real-world autonomous flight tests (waypoint navigation, obstacle avoidance, return-to-launch) — achieving 18 minutes of stable flight endurance at 1.4 kg total weight.
- Developed and validated during a period of active regional conflict in Lebanon (2024–2025), which caused severe GNSS/network disruption and restricted outdoor UAV flight testing — requiring extensive simulation-based validation prior to constrained real-world flight windows.
- Formally proposed to the Lebanese Civil Defence for operational collaboration via official correspondence — attended by their representative at the FYP defence, who confirmed the system's indispensable value for Lebanon's wildfire crisis.
- Full project documentation, architecture, and results: github.com/wadih553/autonomous-uav-disaster-recon

TECHNICAL SKILLS

- **Robotics & Navigation:** ROS1/ROS2, State Estimation & Multi-Sensor Fusion, Autonomous Navigation, SLAM (research interest), Computer Vision
- **Programming:** C++, Python, MATLAB, Bash/Linux
- **AI / ML:** Deep Learning, Machine Learning, Computer Vision (OpenCV, YOLOv8)
- **Hardware:** Raspberry Pi, Arduino, Microcontrollers, IoT, VHDL
- **Design & Simulation:** AutoCAD, LabVIEW, 3D Mechanical Design (CAD), RViz, QGroundControl/Mission Planner
- **Automation:** PLC, VFD, SCADA, HMI, Industrial Automation

- **Networking:** Cisco CCNA

WORK EXPERIENCE

Process Engineer *Jun 2025 – Present*

Multilane Inc. — Houmal, Lebanon

- Develop Python algorithms for optical alignment and Active Electrical Cable (AEC) production for high-speed data centre interconnects.
- Design and implement PLC+VFD automated control systems for high-precision photonics assembly lines; automate pick-and-place processes using semiconductor industry machinery.

Mechatronics Engineer Intern *Nov 2024 – Feb 2025*

Raymond Feghali Co. (RTF) — Roumieh, Lebanon

- Programmed and troubleshot PLC-based control systems for industrial automation cells; integrated and tuned VFDs for motor/conveyor control.

Security Systems Engineer Intern *Aug 2022 – Oct 2022*

La Maison De La Sécurité S.A.R.L. — Mansourieh, Lebanon

- Installed, configured, and maintained surveillance and alarm systems for large-scale infrastructure; commissioned new stations on the shop floor.

RECOGNITION & IMPACT

- Lebanese Civil Defence — Formal Collaboration Initiative: Wadih and his FYP team formally contacted the Lebanese Civil Defence (info@civildefense.gov.lb), cc'ing university supervisor Dr. Roy Abi Zeid Daou, to propose operational collaboration on the UAV system for fire reconnaissance and victim localization. A Civil Defence representative attended the FYP defence and described the system as “very interesting and indispensable” for Lebanon's wildfire crisis — stating the only barrier to real-world deployment was the absence of institutional funding, not the engineering. Supporting correspondence and documentation available upon request.

ADDITIONAL PROJECTS & ACHIEVEMENTS

- NASA International Space Apps Challenge 2025 — participant, certificate received.
- **Doggo Series — Autonomous Robots:** designed firefighter-assist and sumo competition robots using BLE and MATLAB.
- **Vehicle Conversion to Level 3 Autonomy — Hybrid Systems:** designed hybrid motor-generator systems for enhanced efficiency and autonomous vehicle control.

CERTIFICATIONS & LANGUAGES

- Cisco Certified Network Associate (CCNA).
- Conversion of Vehicles to Level 3 Autonomy Using Hybrid Systems.
- Languages: Arabic — Native | English — Professional Proficiency | French — Bilingual.

REFERENCES

- Prof. Elie Al Ahmar, Dean, La Sagesse University — elie.alahmar@uls.edu.lb
- Dr. Roy Abi Zeid Daou, FYP Supervisor & Mechatronics Chair, La Sagesse University — roy.abizeiddaou@prof.uls.edu.lb
- Mr. Ramy Korkis, UAV & Autonomous Systems Industry — rkkanaan@outlook.com